PRD: Checkpoint Deadlines, Escalation & Approval

Oct 7, 2026 ·

Overview

This iteration makes open checkpoints time-bound: each has an open-day limit, overdue items escalate up the hierarchy, and a developer's fix only becomes compliant after Manager approval, with the approval captured as evidence.

Problem. Today a failed or open checkpoint just changes status. Nobody owns the clock, nobody is chased when it slips, and a fix is never confirmed by anyone, so auditors see open items with no trail.

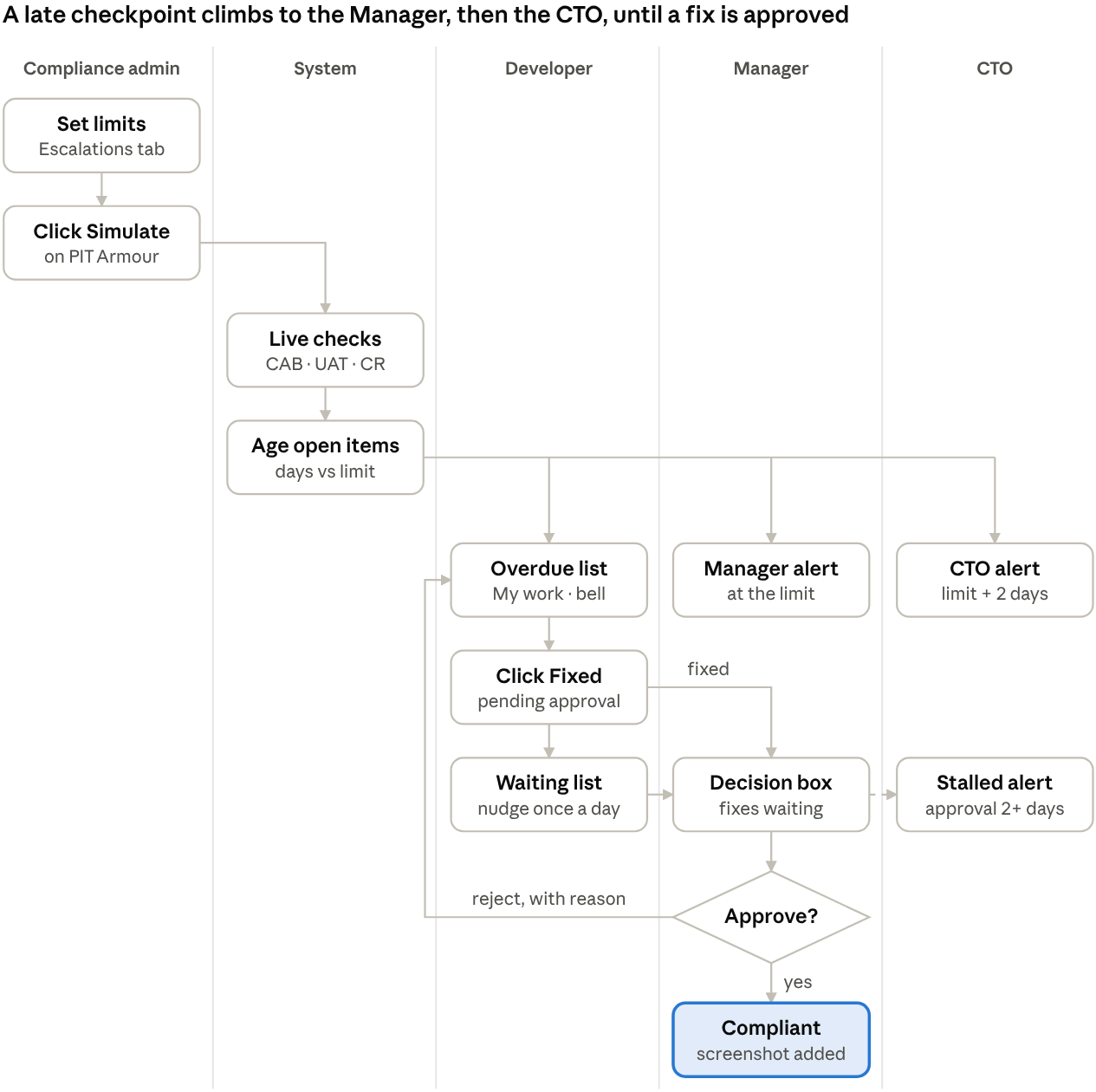
Goal. For every open checkpoint, the right person knows it is late, the next level up hears about it automatically, and closure is approved and evidenced.

Scope of this iteration (POC).

- Project: **PIT Armour** only, with seeded demo data. Other projects (SAATHI, MPS) use the default template and are wired the same way later.
- Trigger: the existing **Simulate** button on the PIT Armour Controls page runs the escalation check on demand. A scheduled daily run comes later.
- Channels: notifications on each stakeholder's dashboard (bell, My work, Decisions box). Email is an extra channel, switched on when recipient addresses are set.
- Built on what exists: role views (CTO, Manager, Developer), invite-link developer access, Simulate, Decisions inbox, Waiting on others with Nudge, and the screenshot evidence agent.

Source. Meeting takeaways (Akhil, Sanket, Pravin, Anushka): checkpoint deadlines and tags, the configuration form, the escalation matrix, developer resolution with manager approval, and the quarterly view. The quarterly filter is tracked separately.

End-to-end flow



escalation and approval flow · 5 roles, 1 decision

Simulate ages every open item against its limit and alerts each level that is due; the developer's Fixed sends it to the Manager, and only an approval makes it compliant. A stalled approval (dashed) alerts the CTO; a rejection loops back to the developer.

Personas and permissions

Four roles touch this flow; each sees only what it must act on.

Role	Demo persona	Sees	Can do	Escalations tab
Developer	Rahul Verma (via invite link)	My work: overdue and escalated items, Waiting on others, bell	Mark **Fixed**, **Nudge** (once a day per item)	No access
Manager	Neha Gupta	Decisions box (fixes waiting, nudges), Escalations, bell	**Approve**, **Reject** with a one-line reason	View
CTO	CTO (name to confirm)	Alerts for items 4+ days open and stalled approvals	Follow up; no approval step in this iteration	Edit
Compliance admin	Priya Rao (current admin login)	Everything, incl. Simulate on PIT Armour	Run Simulate, edit limits and the matrix	Edit

The View-as dropdown (CTO, Manager, Developer) stays the way to show each role in a demo; an invited developer is locked to the Developer view of PIT Armour.

Escalations tab

A dedicated **Escalations** tab holds every project's open-day limits and its escalation matrix; onboarding only picks a template.

Navigation. A new sidebar item, Escalations. It opens on **project folders** (PIT Armour, SAATHI, MPS, matching the Projects page), each with a chip: "Default template" or "N custom limits".

Inside a project, top to bottom:

1. **Escalation matrix**, two stages:

Stage	Who acts	Limit	Escalates to
Fixing the issue	Developer (checkpoint owner)	Per checkpoint, default 2 days	Manager at the limit, CTO 2 days after the limit

Stage	Who acts	Limit	Escalates to
Approving the fix	Manager	2 days per project (editable)	CTO

2. **People per role**: name and email for Developer, Manager and CTO. Email can be blank; notifications still show on dashboards.
3. **Checkpoint limits**: all checkpoints grouped by category in **collapsible folders**, never one long list. Each category shows one limit (e.g. "Change Management · 2 days") that applies to all its rows; expand to override a single checkpoint. Search jumps to a checkpoint by name.

Inheritance. Org default template → project override → checkpoint override. Each row is labelled *default* or *custom*, with **Reset to default**. Only differences from the level above are stored.

Onboarding. The Add Project wizard gains one choice: "Deadline policy: standard template", with a link to customise later in this tab.

Editing. Changes apply on the next Simulate run. Show a short "Saved" confirmation; no approval step for config changes in this iteration.

Simulate: escalation run

Simulate on PIT Armour first runs the existing live checks, then ages every open checkpoint against its limit and notifies each level that is due.

Step A, verify (exists today). CAB #10, UAT #38 and CR #18 run and update their status.

Step B, escalate (new). For each checkpoint still open (non-compliant, needs review or manual in progress):

- Age = today minus its opened date, in calendar days.
- Shown as three numbers: "Open 4 days · limit 2 days · 2 days overdue".

Condition	Notified	Where
Within limit, 1 day left	Developer	Developer bell, My work
Age ≥ limit	Manager	Manager bell, Escalations; email if set

Condition	Notified	Where
Age ⩾ limit + 2 days	CTO (Manager in CC)	CTO alerts; email if set
Fix pending approval ⩾ 2 days	CTO	CTO alerts; email if set

Every run notifies again, by design, so the POC can be shown many times. Email subjects carry the run time, e.g. "Escalation · PIT Armour · 2 overdue checkpoints · 14:32".

Result panel in the Simulate dropdown: the live-check answers as today, then "Escalations: Developer 1 · Manager 2 · CTO 1", each with a preview and *Sent* or *Not sent: reason* for email.

Email (optional channel). One email per recipient per run, listing their checkpoints with owner and age. Sent via Microsoft Graph when the mailbox settings and Mail.Send permission are in place; otherwise only dashboard notifications appear.

Developer and Manager actions

The developer closes their part with one **Fixed** click; only the Manager's approval makes a checkpoint compliant.

Developer (My work)

- New section **Overdue & escalated**: each item shows checkpoint, project, "Open N days · limit L · X days overdue", and a badge for the level reached (*Escalated to Manager*, *Escalated to CTO*).
- **Fixed** button. One click, no form in this iteration. Status becomes **Pending Manager approval**; the developer's clock stops, so the item no longer counts as their overdue.
- The item moves to **Waiting on others** as "Waiting for Neha (N days)", with **Nudge**.
- **Nudge**: notifies the Manager on the dashboard and by email if set. Limited to once a day per item; the button then reads "Nudged · available tomorrow".
- Bell and Open actions KPI update immediately.

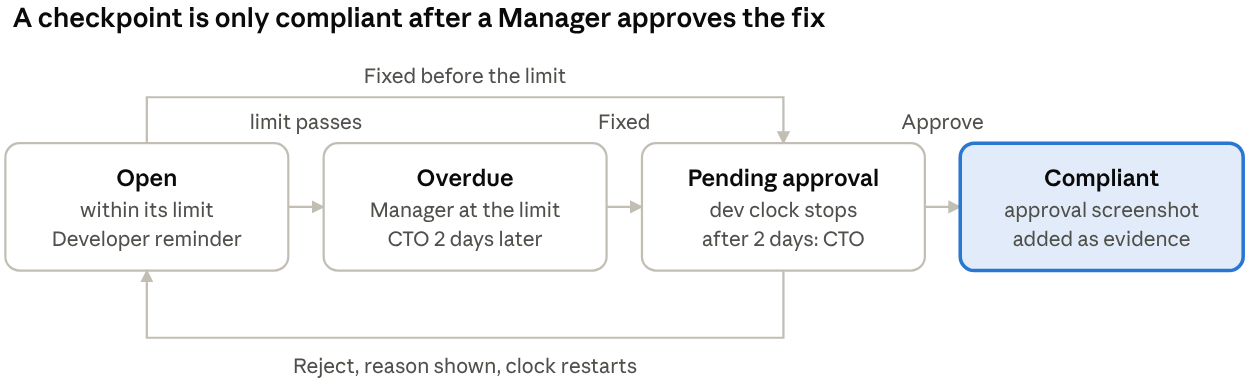
Manager (Decisions box)

- New group **Fixes waiting for your approval**: checkpoint, developer, when fixed, how overdue it was ("Was 2 days overdue · fixed today"), and a **Nudged ×N · last 1h ago** badge.
- **Approve** → status **Compliant**; the item leaves both dashboards.
- **Reject** → a one-line reason is required; the item returns to the developer as **Open**, with the reason shown, and its escalation clock restarts from the developer.
- The Manager bell and the "What needs my decision?" KPI include these items.
- **Approval limit**: 2 days (from the matrix). Past it, the CTO is notified on the next Simulate run.

Evidence on approval

On **Approve**, the screenshot agent renders an **approval record card** and attaches it as the checkpoint's evidence (reusing the CP10/CP38 HTML-to-PNG capture). The card shows checkpoint and project, who fixed it and when, how overdue it was, nudges, approved by and timestamp, and the short history. It opens from **View evidence** on the checkpoint card and appears in the audit report.

Status lifecycle and audit trail



checkpoint status lifecycle · 4 states

Fixed can happen before or after the limit; either way the item waits in Pending approval until the Manager decides.

Every step writes a timestamped history event, shown on the checkpoint card, the Activity page and in the audit report.

Event	Recorded when	Shown in
Escalated (Developer, Manager or CTO)	A Simulate run notifies that level	History, Activity, the recipient's bell
Email sent / not sent	Per recipient, per run	History, Simulate result panel
Marked fixed	Developer clicks Fixed	History, Activity, Manager Decisions box
Nudged	Developer nudges (once a day)	History, Manager Decisions box
Approved	Manager approves; screenshot attached	History, Activity, evidence, audit report
Rejected (with reason)	Manager rejects	History, Developer's list
Approval stalled	Pending over 2 days on a run	History, CTO alerts
Limit or matrix changed	Saved in the Escalations tab	Activity

Demo seed data (PIT Armour)

PIT Armour ships pre-filled so a demo needs no setup: open PIT Armour, click Simulate, and every level gets notified.

Dates are stored **relative to today** ("opened 4 days ago") and resolved on load, so the demo looks the same on any day.

Checkpoint (example)	Opened	Limit	Shows as	Notifies
#15 Rollback plan documented	4 days ago	2 days	2 days overdue	Manager + CTO
#21 PR review before merge	4 days ago	2 days	2 days overdue	Manager + CTO
#8 Impact analysis before implementation	2 days ago	2 days	due today	Manager

Checkpoint (example)	Opened	Limit	Shows as	Notifies
#19 Branching strategy followed	1 day ago	2 days	1 day left	Developer reminder
#6 Change linked to an approved CR (fixed)	fixed 3 days ago, pending approval	approval 2 days	approval 1 day overdue	CTO

All other PIT Armour checkpoints use the default template (2 days). SAATHI and MPS show "Using default template" with no seeded items.

People: Developer Rahul Verma, Manager Neha Gupta, CTO to confirm; emails added when available.

Backend and data model

The backend owns all shared state so the Developer (invite link, own browser), Manager and CTO see the same data. Storage follows the existing invites pattern: JSON files in the FastAPI service for the POC, a database later.

Data

Record	Key fields
EscalationPolicy	projectId, fixStage {managerAtLimit, ctoAfterDays: 2}, approvalStage {limitDays: 2}, people {developer, manager, cto: name + email}
CheckpointLimit	projectId, category or checkpointId, limitDays, source (default / project / checkpoint)
OpenItem	projectId, checkpointId, ownerRole, openedAt, status, fixedAt, fixedBy, approvedAt, approvedBy, rejectReason, lastNudgeAt, nudgeCount
Notification	id, projectId, checkpointId, recipientRole, level (developer / manager / cto / approval), runAt, emailStatus
HistoryEvent	checkpointId, type, actor, at, detail

API (new)

Method + path	Purpose
GET / PUT /api/escalations/policy/{projectId}	Read or save the matrix and people
GET / PUT /api/escalations/limits/{projectId}	Read or save limits with inheritance resolved
POST /api/escalations/run/{projectId}	Simulate step B: age items, create notifications, send email if configured
GET /api/notifications?role=&projectId=	Feed for each dashboard's bell and sections
POST /api/items/{id}/fixed	Developer marks fixed
POST /api/items/{id}/nudge	Nudge, once a day per item
POST /api/items/{id}/approve and /reject	Manager decision; approve triggers the evidence capture
POST /api/demo/seed/{projectId}	Reload PIT Armour's relative-date seed

Email reuses the invite code's Microsoft Graph sendMail. The approval evidence reuses evidence_screenshot.py.

Build plan, acceptance and open questions

Build in seven steps, each demoable on its own; the frontend runs on demo data until the backend steps land.

1. Escalations tab with PIT Armour demo values (frontend only).
2. Simulate step B: ageing and dashboard notifications.
3. Developer: Overdue & escalated, Fixed, Waiting on others with Nudge.
4. Manager: Fixes waiting for approval, Approve and Reject.
5. Backend storage and APIs, so all three dashboards share data across browsers.
6. Approval evidence screenshot.
7. Email, once recipient addresses are set.

Acceptance criteria

- [ ] Clicking Simulate on PIT Armour shows Developer, Manager and CTO notifications matching the seed table.

- [ ] Changing a limit in the Escalations tab changes who is notified on the next run.
- [ ] Fixed moves an item to Waiting on others and to the Manager's Decisions box, in another browser too.
- [ ] Nudge shows on the Manager dashboard and is limited to once a day per item.
- [ ] Approve makes the checkpoint Compliant and attaches the approval screenshot as evidence.
- [ ] Reject returns the item to the developer with the reason.
- [ ] Every step appears in the checkpoint history and the Activity page.

Out of scope this iteration

- Fix form with note and proof (Fixed stays a single button).
- Scheduled daily escalation run (Simulate triggers it).
- Escalation levels beyond the CTO; per-checkpoint approval limits.
- Quarterly filter next to Simulate (separate pointer).
- Business-day calendars and holidays (calendar days for now).

Open questions

- Recipient emails for Developer, Manager and CTO, and the CTO's name.
- Master checkpoint count: 84 (meeting), 85 (dashboard) or 86 (Excel).
- Which roles may edit the Escalations tab: CTO and Compliance admin only?